

Cooperative Load Transportation Using Three Quadrotors

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Abstract—Unmanned aerial vehicles (UAV) provide a wide range of applications, amongst which is load transportation. However, they have low payload capabilities, in general. Then, whenever a heavier load should be transported it is necessary to use a team of UAVs to accomplish the task. In this work, a team of three UAVs in a formation are used to carry a cargo heavier than the capacity of a single vehicle. A nonlinear control strategy based on feedback linearization is used to guide each aircraft in positioning and trajectory tracking tasks while keeping the formation stable, compensating the disturbances caused by the load and the cross effect of cable tensions.

I. INTRODUCTION

Nowadays, unmanned aerial vehicles (UAV), popularly known as drones, are a reality. In recent years, there has been a growing selling of civilian multi-rotor vehicles for recreational use, most of them having four propellers attached to a rigid body (known as quadrotors). These vehicles have a great capacity of three-dimensional flight, they can take-off and land vertically, change its direction and land without any lateral movement, besides moving longitudinally or laterally, or even combining such movements.

These characteristics have made them a sales success and suitable to be used in a variety of tasks, such as management of natural hazards [1], [2], public safety [3], intervention in hostile environments [2], [4], search and rescue missions [5], agriculture [6] and load transportation [7], for instance.

These vehicles have six degrees of freedom (DOF), but just four actuators to control them (in the case of a quadrotor, each one of the four propellers). These properties characterize them as a kind of vehicle known as an underactuated system, meaning that they have more variables to control than input signals. Thus, a reliable model and controller are hard to achieve. Indeed, several techniques have been applied, leading to a diversity of possibilities to guide these kind of vehicles [8]–[13].

Considering cargo transportation, one can use grasps to carry the load attached to the UAV body, what increases its inertia and slows down the attitude response [14]–[16]. Also, to take and deliver a load, the vehicle should land completely and take off afterwards, what can be dangerous sometimes. Another drawback is the shift in the UAV center of mass, once the load will always be attached in a different way [14].

Another approach is the use of cables, what neither does interfere with the attitude response nor changes the center of gravity of the vehicle. However, it adds more underactuated degrees of freedom due to the load oscillations, controlled indirectly by the four propellers. As a drawback, the load oscillations affects the UAV movements and stability, so that it can be led to a crash. However, such oscillations can either be avoided using a smoother or swing-free trajectory, or it can be attenuated using an appropriate control technique in connection with a reliable UAV-load model, preventing accidents and assuring a smooth and safe flight [17], [18].

In [19], for instance, Euler-Lagrange equations are used to obtain the UAV-load dynamic model, together with a passivity based controller, which incorporates a total energy-shaping to guide the system through trajectory-tracking tasks aiming at minimizing load oscillations. The load swings in [20] are reduced using an H_∞ controller, considering the actuated and non-actuated DOF. Newton-Euler equations are used in [21], with a backstepping controller with integral action to guide the UAV while suppressing the load disturbance. The same model technique is used in [22], now with a geometric controller, as in [23], where attention is given to the load lift task, which induces a nonlinear behavior of the vehicle when the load loses contact with the ground. An approximation of a flexible cable is presented in [18], whereas in [24] there is an extension of this work regarding cooperation involving multiple vehicles.

Another meaningful aspect is that the use of a larger and more specialized vehicle may be more expensive than the use of several smaller ones. In fact, the use of multiple vehicles may increase the task capability and performance, but also overcome individual limitations, as low payload capacity, in load transportation applications. However, it increases the system complexity, because a formation control should be considered for assuring a coordinated flight [25], [26].

In [5] an autonomous miniature helicopter is used for aerial transportation and deployment, as part of a multi-UAV system composed by three small-sized helicopters and a heavy payload, which cannot be carried by a single vehicle. Torque and force sensors are used in the cables to feedback the control loop. In [27] an iterative LQR is used to control a team of two quadrotors, whereas in [28] a passivity-based controller is used for a team of three UAVs.

In [29] a team of UAVs carry a rigid body while in [24] flexible cables are used with a geometric controller, similar to [30]. When the number of vehicles carrying a load increases from one to two, it bounds the load in the plane composed by the two vehicles [31], allowing oscillations only in the orthogonal plane. Using three or more vehicles, the load is

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confined by the cables in multiple planes, which limits the load movement, not allowing free oscillations. Moreover, the load displacement will be a result of the formation composed by the vehicles, which are physically attached to it.

Inserted in the aforementioned context, this work proposes a coordinated aerial transportation, where a team of three quadrotors transports a point mass connected by cables. Here, the problem of transporting heavy payloads is addressed (such as in package delivery), where one vehicle alone is not able to lift the load. The load position is controlled by the vehicles to accomplish positioning and trajectory tracking tasks using a nonlinear control strategy based on feedback linearization. Numerical experiments are presented and discussed, which allow concluding that the proposed method is efficient in accomplishing the tasks addressed. Notice that this is an extension of the work presented in [31], where two quadrotors carry a load suspended by cables.

To make easier to follow the ideas addressed, the paper is hereinafter split in a few sections, beginning with Section II, which presents a short description of the UAV underactuated model. In the sequel, Section III presents the formation equations and how its movement affects the load position as well as their effect in each vehicle, whereas Section V presents the results using this formation with a nonlinear controller in positioning and trajectory tracking tasks. Finally, VI highlights the main conclusions of the work.

II. DYNAMIC MODEL

A quadrotor has four propellers, each one generating a force $f_i(t)$, $i = 1, 2, 3, 4$, in the direction perpendicular to the vehicle platform. The vehicle movement is performed through combining these forces. In this work, the mathematical modeling is based on the Parrot AR.Drone 2.0, that has an important detail: this frame is not aligned with the $^b x$ and $^b y$ axes; instead, it is rotated 45° around the $^b z$ axis. Thus, lateral and longitudinal movements are performed by suitably controlling the rotation of all the four motors (See Figure 1) [17], [32].

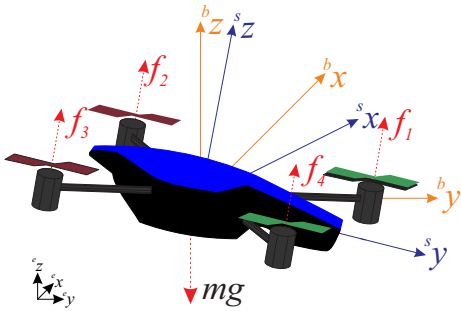


Fig. 1: Parrot AR.Drone 2.0[®] quadrotor, its frame alignment and propulsion forces.

The pose of each quadrotor in the space is defined as $\mathbf{q} = [\xi_i \ \eta_i]^T$, according to the inertial reference system $\langle e \rangle$, where i corresponds to the i -th body, being $i = 0$ for the load, and $i = 1, 2, 3$ for each quadrotor. $^e \xi_i = [x_i \ y_i \ z_i]^T \in \mathbb{R}^3$ correspond to longitudinal, lateral and

normal displacements, whereas $\eta = [\phi \ \theta \ \psi]^T \in \mathbb{R}^3$, gives the pitch, roll and yaw angles, respectively, related to the spatial reference system $\langle s \rangle$, whose origin coincides with the center of mass of each body. Notice that the load, as it is considered a point-mass, gets confined between the cables, not having attitude angles: its orientation is defined by the formation, so that $\mathbf{q}_0 = \xi_0$ where η_0 is omitted because $\eta_0 = \mathbf{0}$. Figure 2 illustrates the reference system and the variables involved. It is worth mentioning that because the load is a rigid body its attitude should be considered.

The dynamic model of a quadrotor, in the underactuated form, in the inertial frame $\langle e \rangle$ is given by [33]

$$\begin{bmatrix} mI_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_r \end{bmatrix} \begin{bmatrix} \ddot{\xi} \\ \ddot{\eta} \end{bmatrix} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_r \end{bmatrix} \begin{bmatrix} \dot{\xi} \\ \dot{\eta} \end{bmatrix} + \begin{bmatrix} \mathbf{G} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{D}_t \\ \mathbf{D}_r \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \boldsymbol{\tau} \end{bmatrix}, \quad (1)$$

where $I_3 \in \mathbb{R}^{3 \times 3}$ is the identity matrix, whereas $\mathbf{M}_r(\eta) = W_\eta^T \mathbf{J} W_\eta \in \mathbb{R}^{3 \times 3}$ is the rotational one. By their turn, $\mathbf{C}_r(\eta, \dot{\eta}) = \dot{\mathbf{M}}_r - \frac{1}{2} \dot{\eta}^T \frac{\partial \mathbf{M}_r}{\partial \eta} \in \mathbb{R}^{3 \times 3}$ are the matrices of Coriolis and centrifugal forces, and $\mathbf{G} = [0 \ 0 \ mg]^T$ is the vector of gravity force. Finally, $\mathbf{D}_t$ and $\mathbf{D}_r$ are the load translational and rotational effects, respectively.

The suspended load does not produce any rotational disturbance on the vehicles to which it is connected. Once confined, its movements depend on the position of each UAV in the formation and not how they are oriented. Thus, $\mathbf{D}_r = \mathbf{0}$.

III. FORMATION DESCRIPTION

When a load is carried by a single vehicle, it behaves as a pendulum in the free space [17]. When the load is carried by two vehicles, by its turn, their oscillations in one plane are restricted, since there are cables in opposite directions pulling it [31]. In the present case, as the load is confined by three cables, evenly distributed in the space, any oscillation is contained, so that it does not have the behavior of a pendulum anymore, although still affecting how the vehicles fly according to its relative position.

It can be considered that the vehicles are indirectly connected one to the two others (see Figures 2 and 3). This means that the movement of one UAV changes the formation variables, leading to changes in the load position, once the cables have a fixed length. This shift alters the forces decomposition, changing the force applied to the load and each one of the vehicles, causing a cross effect of the cable tensions, where one vehicle affects its neighbors. This way the distance among the vehicles defines how the forces interact with them. Such distances are calculated as

$$S_f = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \\ \sqrt{(x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2} \\ \sqrt{(x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2} \end{bmatrix}, \quad (2)$$

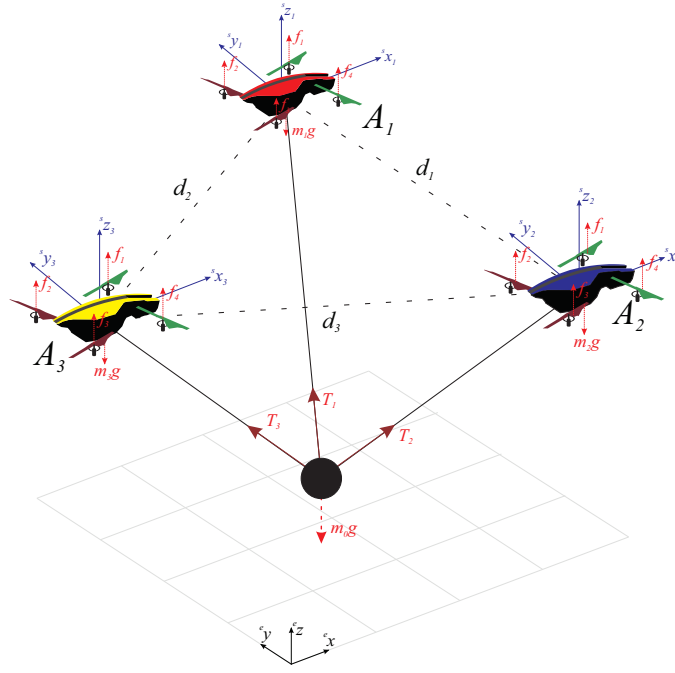


Fig. 2: The three Parrot AR.Drone 2.0® quadrotors carrying a cable-suspended load.

whereas the angles between adjacent distances are given by

$$\beta_i = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} = \begin{bmatrix} \arccos \frac{d_1^2 + d_2^2 - d_3^2}{2d_1d_2} \\ \arccos \frac{d_1^2 + d_3^2 - d_2^2}{2d_1d_3} \\ \arccos \frac{d_2^2 + d_3^2 - d_1^2}{2d_2d_3} \end{bmatrix}. \quad (3)$$

The centroid of the triangle formed by the quadrotors is considered as the formation center, where $\mathbf{P}_F = (\xi_1 + \xi_2 + \xi_3)/3$. Notice that the load position is congruent with x_F and y_F , so that $x_0 = x_F$ and $y_0 = y_F$. The direction of this point, using as reference each UAV, is named α_i , which

is the angle between its neighbor and the formation central point, where

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} \arccos \frac{d_1^2 + d_2^2 - d_3/4}{2d_1d_2} \\ \arccos \frac{d_1^2 + d_3^2 - d_2/4}{2d_1d_3} \\ \arccos \frac{d_2^2 + d_3^2 - d_1/4}{2d_2d_3} \end{bmatrix}. \quad (4)$$

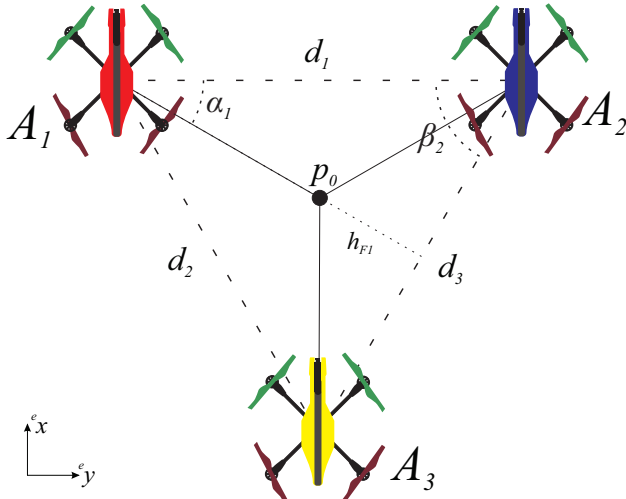


Fig. 3: A top view of the three quadrotors carrying a load.

Finally, $h_{F1} = \sqrt{\frac{1}{2} \left(d_1^2 + d_2^2 - \frac{1}{2} d_3^2 \right)}$ is the distance between A_1 and the central point of the opposite segment, passing through (x_F, y_F, z_F) .

Using these values and analyzing the front view of the formation, as shown in Figure 4, one can determine the height between load and the line connecting the UAVs A_1 and A_2 of the formation, with $h_0 = l' \sin \gamma_1$, with $l'_i = l \cos(\alpha_i)$, where l'_i is the projection of the cable length l in the plane yz .

The distances d_i and the angles α_i and β_i determine how each vehicle affects its neighbor. In fact, the greater the distance d_i , the greater the force pulling the vehicles is. It is considered the situation when the vehicles are at the same distance one from each other, meaning that $d_1 = d_2 = d_3$, which leads to $\beta_1 = \beta_2 = \beta_3 = 60^\circ$, $\alpha = 30^\circ$ and $\gamma = 45^\circ$. With these desired values, the power consumption of the vehicles will be equivalent. Also, notice that these distances are relative to the cable length l , so that a suitable cable length should be chosen.

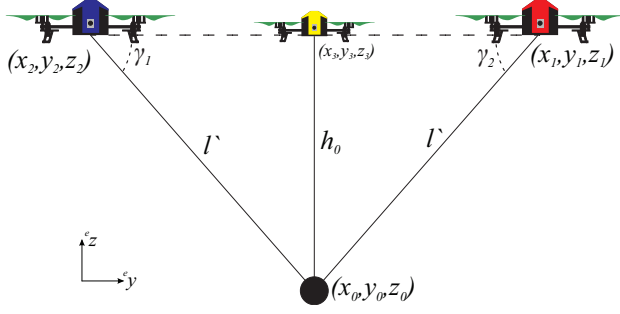


Fig. 4: Frontal view of the formation of three quadrotors carrying a cable-suspended load.

IV. THE NON-LINEAR CONTROLLER

To guide each vehicle throughout the flight tasks, it is used a nonlinear controller based on feedback linearization. To design such a controller, the dynamic model of the UAV is obtained using Euler-Lagrange equations. After that, the relationship obtained is separated into two parts, regarding the active (index a) and the passive (index p) variables, as it is done in [33], thus obtaining

$$\mathbf{F} = \mathbf{M}_{ap}\ddot{\mathbf{q}}_p + \mathbf{M}_{aa}\ddot{\mathbf{q}}_a + \mathbf{E}_a \quad (5)$$

and

$$\mathbf{0} = \mathbf{M}_{pp}\ddot{\mathbf{q}}_p + \mathbf{M}_{pa}\ddot{\mathbf{q}}_a + \mathbf{E}_p, \quad (6)$$

where $\mathbf{E}_a = \mathbf{D}_a + \mathbf{G}_a$ and $\mathbf{E}_p = \mathbf{D}_p + \mathbf{G}_p$.

Isolating $\ddot{\mathbf{q}}_p$ in (6), it comes that

$$\ddot{\mathbf{q}}_p = \mathbf{M}_{pp}^{-1}\mathbf{M}_{pa}\ddot{\mathbf{q}}_a - \mathbf{M}_{pp}^{-1}\mathbf{E}_p. \quad (7)$$

Then, replacing (7) in (5) one gets

$$\begin{aligned} \mathbf{F} &= -\mathbf{M}_{ap}\mathbf{M}_{pp}^{-1}(\mathbf{M}_{pa}\ddot{\mathbf{q}}_a + \mathbf{E}_p) + \mathbf{M}_{aa}\ddot{\mathbf{q}}_a + \mathbf{E}_a, \\ &= \underbrace{\mathbf{M}_{aa} - \mathbf{M}_{ap}\mathbf{M}_{pp}^{-1}\mathbf{M}_{pa}}_{\bar{\mathbf{M}}_{aa}}\ddot{\mathbf{q}}_a + \underbrace{\mathbf{E}_a - \mathbf{M}_{ap}\mathbf{M}_{pp}^{-1}\mathbf{E}_p}_{\bar{\mathbf{E}}_a}. \end{aligned} \quad (8)$$

Applying partial feedback linearization the control law

$$\mathbf{F} = \bar{\mathbf{M}}_{aa}\boldsymbol{\eta}_a + \bar{\mathbf{E}}_a, \quad (9)$$

is proposed, where $\boldsymbol{\eta}_a = \ddot{\mathbf{q}}_{ad} + \mathbf{K}_{a1}\tanh(\mathbf{K}_{a2}\dot{\tilde{\mathbf{q}}}_a) + \mathbf{K}_{a3}\tanh(\mathbf{K}_{a4}\tilde{\mathbf{q}}_a)$, with $\mathbf{K}_{ai} \in \mathbb{R}^4$ being positive definite diagonal gain matrices.

Therefore, the closed-loop equation is

$$\ddot{\tilde{\mathbf{q}}}_a + \mathbf{K}_{a1}\tanh(\mathbf{K}_{a2}\dot{\tilde{\mathbf{q}}}_a) + \mathbf{K}_{a3}\tanh(\mathbf{K}_{a4}\tilde{\mathbf{q}}_a) = \mathbf{0},$$

which guarantees that $\tilde{\mathbf{q}}$ and $\dot{\tilde{\mathbf{q}}}$ tend to zero asymptotically (see [33] for more details).

V. RESULTS

To validate the proposed method for transporting a suspended load by a formation with three UAVs, this section presents a series of simulations. Firstly a positioning task is presented, and after a trajectory tracking task is considered. The experiments were performed using the *AuRoRA Platform*, a framework created by our research group to deal with simulations and experiments, in which the models and

controller proposed here were implemented (see [34] for details), just like in some other works, such as [31], [32], [35], [36]. The model is based on the Parrot AR.Drone 2.0[®] quadrotor, and has already been validated in [33] and [12].

A. Positioning Task

The results for the first experiment are depicted in Figure 5, where the subscript D represents the desired values of the variables. In this experiment the vehicles should take-off and carry a load through check points. Notice that the load only begins to affect the vehicles after being hoisted off the ground, in this case around instant $t = 6$ s. Moreover, although the paths followed by each robot look similar, they are effectively different, because each vehicle flies in a different way, since the load affects each of them differently, according to its position in the formation.

Notice, also, that the load has a known position in the formation, which is related to the positions of the vehicles. Thus, a desired value can be chosen for the load position, simplifying the cargo delivery, which is the objective to be achieved.

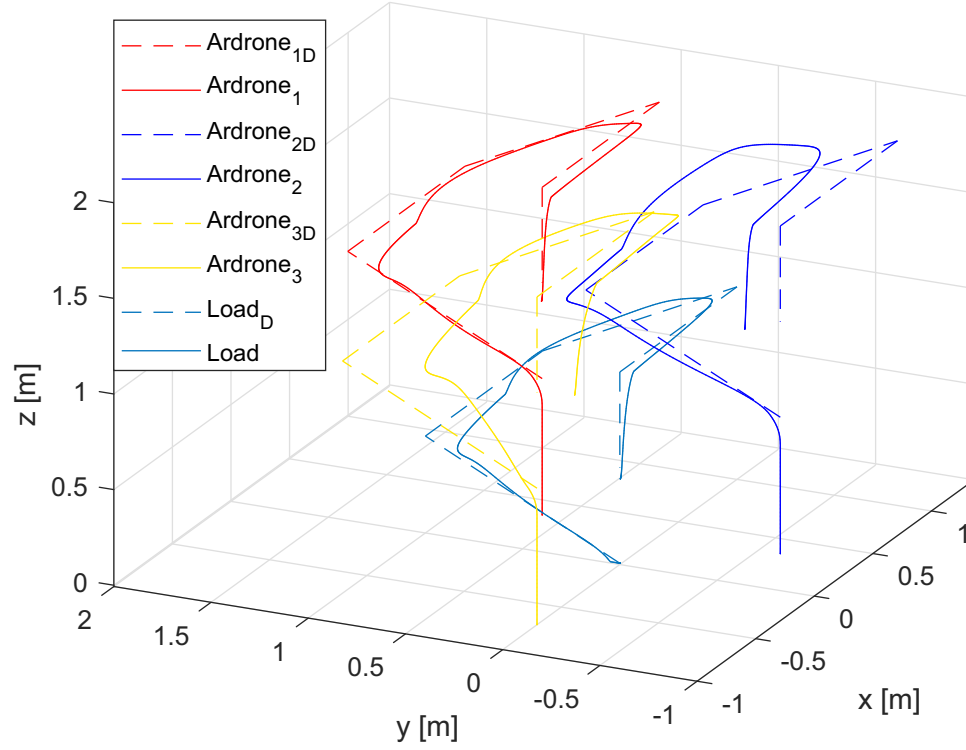
B. Trajectory-Tracking Task

The results correspondent to a trajectory-tracking task are illustrated in Figure 6, where the subscript D also represents the desired values of the variables once more. The trajectory being tracked is an inclined lemniscate curve.

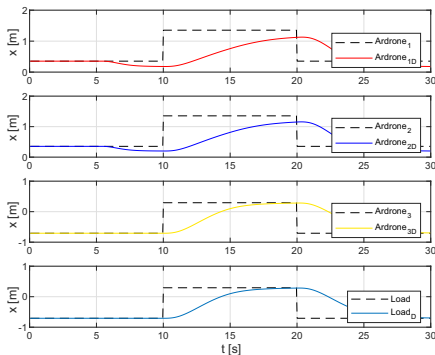
The quadrotors starts the task accomplishment with a take-off maneuver and, in the sequel, after the instant $t = 10$ s, they start following the lemniscate trajectory. Notice again that each vehicle has different errors with respect to the desired values, once the load affects them in different ways, depending on their positions in the formation. However, all the UAVs track the desired trajectory with just small errors with respect to the desired values, as one can see from the figures.

VI. CONCLUDING REMARKS

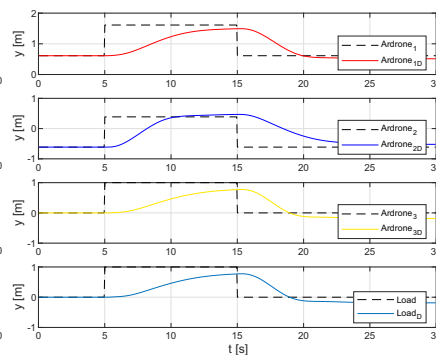
UAVs have opened up a wide range of applications, one of which is load transportation. Unfortunately, the load capacity of these kind of vehicles is still low, and UAVs able to carry heavy loads are too much expensive. The use of multiple vehicles can outcome this deficiency, although increasing the modeling and control complexity. In this work a formation of three quadrotors carrying a heavy load suspended by cables is considered. In this situation the load affects all vehicles, because it is configured a situation in which the force produced by one vehicle is transmitted to the others through the cables attaching the load to them. The contribution was the use of a nonlinear controller based on feedback linearization, designed with basis on the underactuated model based on Lagrange equations of the UAVs-load system, considering these effects and also guiding the vehicles and the load throughout positioning and trajectory-tracking tasks. Numerical experiments presented validate the proposal, emphasizing that the load does not present oscillations, a second benefit of the use of three UAVs to carry the load.



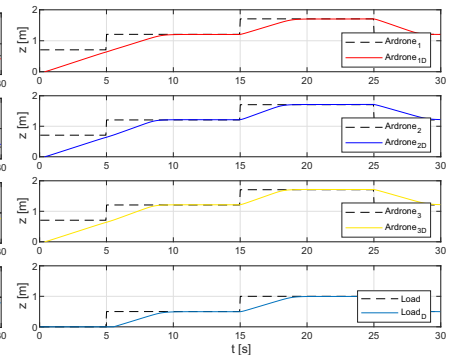
(a) 3D-view.



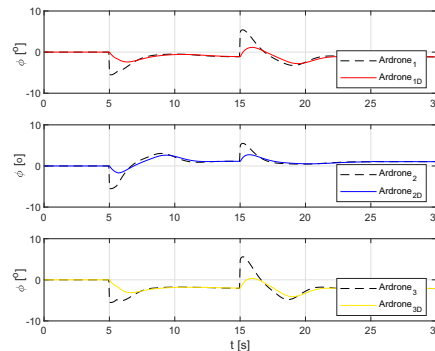
(b)



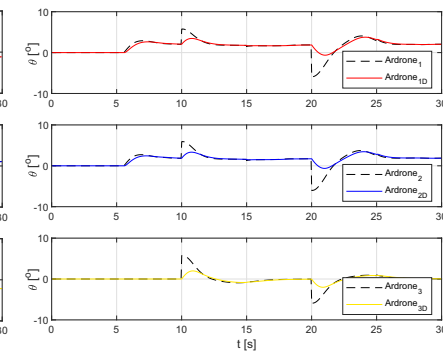
(c)



(d)

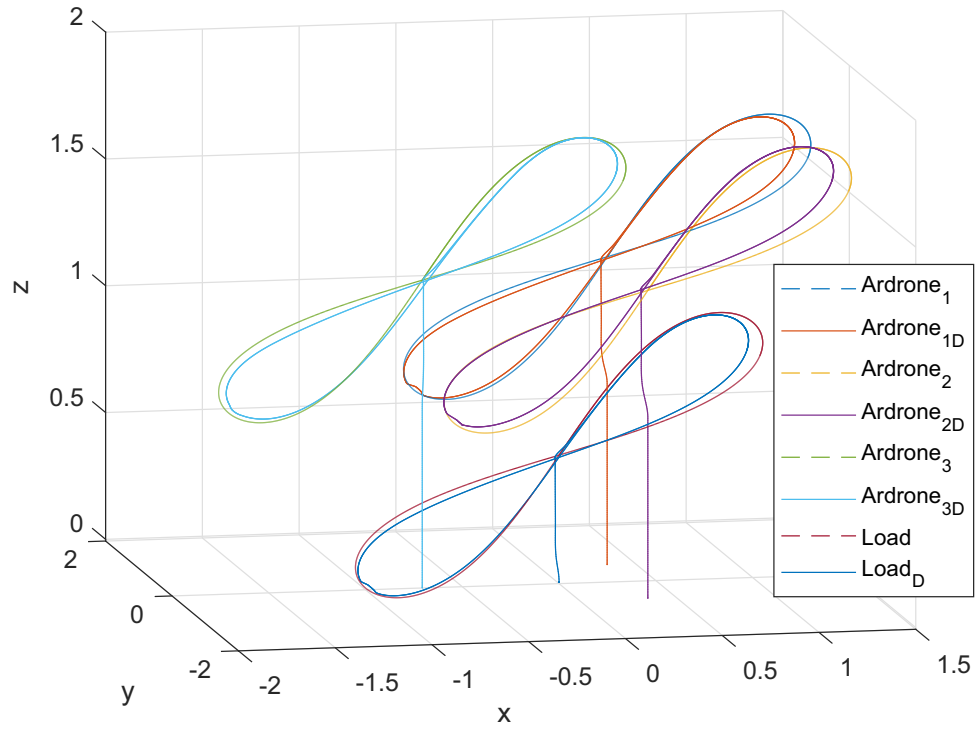


(e)

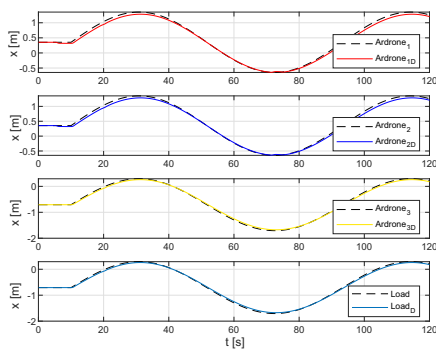


(f)

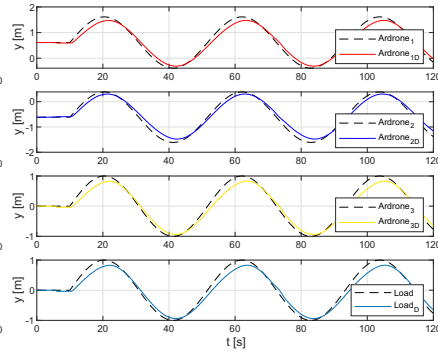
Fig. 5: Results for the positioning task. The 3D view is presented in (a) and the coordinates x (b) y (c) and z (d) are presented together with ϕ (e) and θ (f) angles.



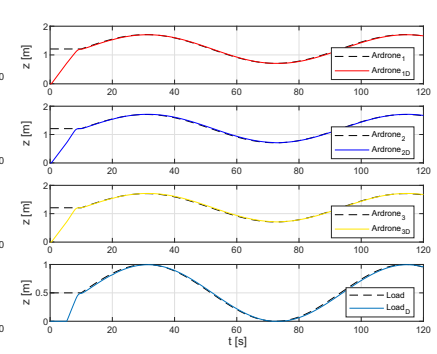
(a) 3D-view.



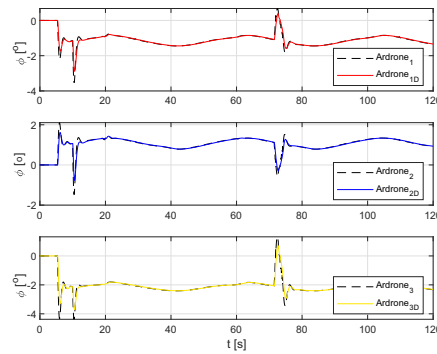
(b)



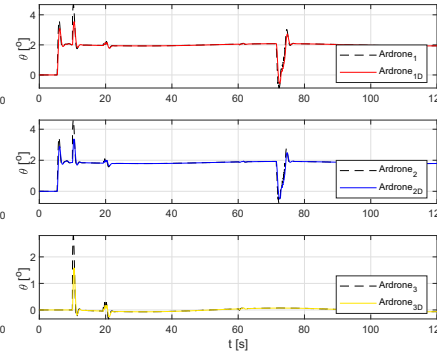
(c)



(d)



(e)



(f)

Fig. 6: Results for the tracking of an inclined lemniscate trajectory. The 3D view is presented in (a) and the coordinates x (b) y (c) and z (d) are presented together with ϕ (e) and θ (f) angles.

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REFERENCES

- [1] I. Maza, F. Caballero, J. Capitán, J. R. Martínez-de Dios, and A. Ollero, "Experimental results in multi-uav coordination for disaster management and civil security applications," *Journal of Intelligent & Robotic Systems*, vol. 61, no. 1, pp. 563–585, Jan 2011.
- [2] C. Yuan, Z. Liu, and Y. Zhang, "Aerial images-based forest fire detection for firefighting using optical remote sensing techniques and unmanned aerial vehicles," *Journal of Intelligent & Robotic Systems*, vol. 88, no. 2-4, pp. 635–654, 2017.
- [3] K. Kanistras, G. Martins, M. J. Rutherford, and K. P. Valavanis, *Handbook of Unmanned Aerial Vehicles*. Dordrecht: Springer Netherlands, 2015, ch. Survey of Unmanned Aerial Vehicles (UAVs) for Traffic Monitoring, pp. 2643–2666.
- [4] L. Merino, J. R. Martínez-de Dios, and A. Ollero, *Handbook of Unmanned Aerial Vehicles*. Dordrecht: Springer Netherlands, 2015, ch. Cooperative Unmanned Aerial Systems for Fire Detection, Monitoring, and Extinguishing, pp. 2693–2722.
- [5] M. Bernard, K. Kondak, I. Maza, and A. Ollero, "Autonomous transportation and deployment with aerial robots for search and rescue missions," *Journal of Field Robotics*, vol. 28, 2011.
- [6] Y. Feng, C. A. Rabbath, and C.-Y. Su, *Handbook of Unmanned Aerial Vehicles*. Dordrecht: Springer Netherlands, 2015, ch. Modeling of a Micro UAV with Slung Payload, pp. 1257–1272.
- [7] Y. Feng, C. Alain Rabbath, and C.-Y. Su, *Modeling of the Dynamics of a Micro UAV with a Single Slung Load*. Springer, 12 2018, pp. 1–19.
- [8] Z. Xu, X. Nian, H. Wang, and Y. Chen, "Robust guaranteed cost tracking control of quadrotor uav with uncertainties," *ISA Transactions*, vol. 69, pp. 157 – 165, 2017.
- [9] O. Mofid and S. Mobayen, "Adaptive sliding mode control for finite-time stability of quad-rotor uavs with parametric uncertainties," *ISA Transactions*, vol. 72, pp. 1 – 14, 2018.
- [10] Z. Song and K. Sun, "Adaptive compensation control for attitude adjustment of quad-rotor unmanned aerial vehicle," *ISA Transactions*, vol. 69, pp. 242 – 255, 2017.
- [11] L. V. Santana, A. S. Brandão, and M. Sarcinelli-Filho, "Navigation and Cooperative Control Using the AR.Drone Quadrotor," *Journal of Intelligent and Robotic Systems: Theory and Applications*, vol. 84, no. 1-4, pp. 327–350, 2016.
- [12] A. S. Brandão, D. Gandolfo, M. Sarcinelli-Filho, and R. Carelli, "Pvtol maneuvers guided by a high-level nonlinear controller applied to a rotorcraft machine," *European Journal of Control*, vol. 20, no. 4, pp. 172 – 179, 2014.
- [13] M. C. P. Santos, C. D. Rosales, J. A. Sarapura, M. Sarcinelli-Filho, and R. Carelli, "An adaptive dynamic controller for quadrotor to perform trajectory tracking tasks," *Journal of Intelligent & Robotic Systems*, vol. 93, no. 1, pp. 5–16, Feb 2019.
- [14] P. E. Pounds and A. M. Dollar, *Aerial Grasping from a Helicopter UAV Platform*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2014, pp. 269–283.
- [15] G. Loianno, V. Spurny, J. Thomas, T. Baca, D. Thakur, D. Hert, R. Penicka, T. Krajnik, A. Zhou, A. Cho, M. Saska, and V. Kumar, "Localization, grasping, and transportation of magnetic objects by a team of mavs in challenging desert-like environments," *IEEE Robotics and Automation Letters*, vol. 3, no. 3, pp. 1576–1583, July 2018.
- [16] R. Bhnemann, D. Schindler, M. Kamel, R. Siegwart, and J. Nieto, "A decentralized multi-agent unmanned aerial system to search, pick up, and relocate objects," in *2017 IEEE International Symposium on Safety, Security and Rescue Robotics (SSRR)*, Oct 2017, pp. 123–128.
- [17] I. H. B. Pizetta, A. S. Brandão, and M. Sarcinelli-Filho, "Modelling and control of a quadrotor carrying a suspended load," in *2015 Workshop on Research, Education and Development of Unmanned Aerial Systems (RED-UAS)*, Nov 2015, pp. 249–257.
- [18] F. A. Goodarzi, D. Lee, and T. Lee, "Geometric control of a quadrotor uav transporting a payload connected via flexible cable," *International Journal of Control, Automation and Systems*, vol. 13, no. 6, pp. 1486–1498, Dec 2015.
- [19] M. E. Guerrero-Sánchez, D. A. Mercado-Ravell, R. Lozano, and C. D. Garca-Beltrán, "Swing-attenuation for a quadrotor transporting a cable-suspended payload," *ISA Transactions*, vol. 68, pp. 433 – 449, 2017.
- [20] G. V. Raffo and M. M. de Almeida, "Nonlinear robust control of a quadrotor uav for load transportation with swing improvement," in *2016 American Control Conference (ACC)*, July 2016, pp. 3156–3162.
- [21] K. Klausen, T. I. Fossen, and T. A. Johansen, "Nonlinear control with swing damping of a multirotor uav with suspended load," *Journal of Intelligent & Robotic Systems*, vol. 88, no. 2, pp. 379–394, Dec 2017.
- [22] S. Tang, V. West, and V. Kumar, "Aggressive flight with suspended payloads using vision-based control," *IEEE Robotics and Automation Letters*, vol. 3, no. 2, pp. 1152–1159, April 2018.
- [23] P. J. Cruz and R. Fierro, "Cable-suspended load lifting by a quadrotor uav: hybrid model, trajectory generation, and control," *Autonomous Robots*, vol. 41, no. 8, pp. 1629–1643, Dec 2017.
- [24] F. Goodarzi and T. Lee, "Stabilization of a rigid body payload with multiple cooperative quadrotors," *Journal of Dynamic Systems, Measurement, and Control*, vol. 138, p. 17, 08 2016.
- [25] G. Loianno and V. Kumar, "Cooperative transportation using small quadrotors using monocular vision and inertial sensing," *IEEE Robotics and Automation Letters*, vol. 3, no. 2, pp. 680–687, April 2018.
- [26] D. Mellinger, M. Shomin, N. Michael, and V. Kumar, *Cooperative Grasping and Transport Using Multiple Quadrotors*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 545–558.
- [27] Y. Alothman, M. Guo, and D. Gu, "Using iterative lqr to control two quadrotors transporting a cable-suspended load," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 4324 – 4329, 2017, 20th IFAC World Congress.
- [28] C. Meissen, K. Klausen, M. Arcak, T. I. Fossen, and A. Packard, "Passivity-based formation control for uavs with a suspended load," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 13 150 – 13 155, 2017, 20th IFAC World Congress.
- [29] N. Michael, J. Fink, and V. Kumar, "Cooperative manipulation and transportation with aerial robots," *Autonomous Robots*, vol. 30, no. 1, pp. 73–86, Jan 2011.
- [30] T. Lee, "Geometric control of quadrotor uavs transporting a cable-suspended rigid body," *IEEE Transactions on Control Systems Technology*, vol. 26, no. 1, pp. 255–264, Jan 2018.
- [31] I. H. B. Pizetta, A. S. Brandão, and M. Sarcinelli-Filho, "Avoiding obstacles in cooperative load transportation," *ISA Transactions*, 2019. [Online]. Available: <http://www.sciencedirect.com/science/article/pii/S0019057819300308>
- [32] I. H. B. Pizetta, A. S. Brandão, and M. Sarcinelli-Filho, "Modelling and control of a pvtol quadrotor carrying a suspended load," in *2015 International Conference on Unmanned Aircraft Systems (ICUAS)*, June 2015, pp. 444–450.
- [33] A. S. Brandão, M. Sarcinelli-Filho, and R. Carelli, "High-level underactuated nonlinear control for rotorcraft machines," in *2013 IEEE International Conference on Mechatronics (ICM)*, Feb 2013, pp. 279–285.
- [34] I. H. B. Pizetta, A. S. Brandão, and M. Sarcinelli-Filho, "A hardware-in-loop platform for rotary-wing unmanned aerial vehicles," *Journal of Intelligent and Robotic Systems*, vol. 84, no. 1, pp. 725–743, 2016.
- [35] —, "Cooperative quadrotors carrying a suspended load," in *2016 International Conference on Unmanned Aircraft Systems (ICUAS)*, June 2016, pp. 1049–1055.
- [36] —, "Control and obstacle avoidance for an uav carrying a load in forestal environments," in *2018 International Conference on Unmanned Aircraft Systems (ICUAS)*, June 2018, pp. 62–67.